Algebraic flux correction for convection–diffusion–reaction: a reproduction of the numerical results of Barrenechea, John & Knobloch (SINUM 2016)

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Abstract

We recompute all numerical results of Section 8 of G. R. Barrenechea, V. John and P. Knobloch, *Analysis of algebraic flux correction schemes*, SIAM J. Numer. Anal. 54 (2016) 2427–2451 [1]: Example 20 with $\varepsilon = 10^{-8}$ and $\varepsilon = 10$ on five grids, Tables 1–11. Of the 252 error values computed (levels 3–8, up to $1.05 \cdot 10^6$ unknowns), 245 agree with the paper in all four printed digits; the remaining seven differ by at most 0.08 %, all on levels 7–8 and of the size explained by the paper's looser stopping tolerance. The reproduction settles three points the paper leaves implicit: Table 1 is computed on grid 5 (as its caption says, not on grid 1 as the text says), the mean in Table 7 is taken over all edges, and the error norm $\|e_h\|_h$ uses $C_a = 1$ with the stabilisation term evaluated at $i_h u - u_h$. The unstructured grid 3, not given in the paper, was recovered exactly from the vector drawing of its Figure 1. The fixed-point iteration with Anderson acceleration used in the paper is, on the non-Delaunay and unstructured grids at $\varepsilon = 10^{-8}$, more than ten times faster than a semismooth Newton method; on grid 3 at level 7 both stagnate at a residual of $1.3 \cdot 10^{-8}$.

Contents

1 Scope and summary **2**

2 Problem, scheme and error measures **2**

3 Grids **3**

4 Nonlinear solvers **4**

5 Results **5**

6 Discussion **9**

6.1 The seven deviations 9

6.2 Three choices made explicit 9

6.3 What the data say about the theory 9

7 Reproducibility **10**

1 Scope and summary

The paper [1] proves, for algebraic flux correction (AFC) schemes with symmetric limiters, existence of solutions, a local discrete maximum principle for Kuzmin's limiter, and an a priori error estimate for the convection–diffusion–reaction equation (Corollary 17). Its Section 8 tests the estimate on one smooth example with eleven tables. We recompute every table with the finite element code `galerkit`; Table S lists the agreement table by table and the cost of the finest solve, Figure 2 shows the two quantities the theory is about, and Section 5 gives every table.

Table S: Agreement with the paper per table: entries computed, entries differing in the printed digits, largest relative deviation from the printed values (the printing rounds by up to $5 \cdot 10^{-4}$), and the nonlinear solve on the finest level (solver and iterations, final Euclidean residual, wall time in minutes on one core).

Table	ε	grid	levels	entries	differ	max dev.	finest: solver, it.	residual	min
1	10^{-8}	5	3–8	24	0	3.7e-04	Newton 1	1e-16	5
2	10^{-8}	1	3–8	24	4	7.9e-04	Anderson 3688	1e-10	95
3	10^{-8}	2	3–7	20	0	3.7e-04	Newton 42	2e-16	3
4	10^{-8}	3	3–7	20	1	2.9e-04	Anderson 3400	1e-08	13
5	10^{-8}	4	3–8	24	1	2.5e-04	Anderson 7066	1e-10	173
6	10^{-8}	5	3–8	24	1	3.0e-04	Anderson 6833	1e-10	168
8	10	1	3–7	20	0	4.1e-04	Newton 2	6e-13	1
9	10	4	3–8	24	0	3.0e-04	Newton 1	2e-11	5
10	10	5	3–8	24	0	3.7e-04	Newton 4	1e-12	8
11	10	5	3–8	24	0	4.2e-04	Newton 2	1e-12	10

The main findings are:

1. **The results are reproduced.** Every printed digit agrees except seven entries, all on the two finest levels: four in Table 2, one each in Tables 4, 5 and 6 (Section 6.1).

2. **Three implicit choices of the paper are identified** (Section 6.2): the grid of Table 1, the averaging set of Table 7, and the precise form of $\|e_h\|_h$.

3. **Solver.** Anderson-accelerated fixed point iteration, as in the paper, is the method of choice for $\varepsilon = 10^{-8}$ on grids 3–5; semismooth Newton is faster everywhere else (Section 4).

4. **What the data say about the theory** (Section 6.3): the $O(h^{1/2})$ estimate of Corollary 17 is attained for solution-independent weights and is pessimistic for Kuzmin's limiter on grid 1; in the diffusion-dominated regime the convergence of $d_h(u_h; i_h u, i_h u)^{1/2}$ is decided by the mesh angles exactly as Remark 18 predicts.

2 Problem, scheme and error measures

Example 20. On $\Omega = (0,1)^2$,

$$-\varepsilon \Delta u + \boldsymbol{b} \cdot \nabla u + cu = g \ \text{ in } \Omega, \qquad u = 0 \ \text{ on } \partial\Omega, \qquad \boldsymbol{b} = (3,2)^T, \ c = 1, \tag{1}$$

with g such that $u(x,y) = 100\, x^2(1-x)^2\, y(1-y)(1-2y)$, for $\varepsilon = 10^{-8}$ (convection dominated) and $\varepsilon = 10$ (diffusion dominated). Here $\sigma_0 = c = 1$.

AFC scheme. Let $A = (a_{ij})_{i,j=1}^N$ be the P_1 Galerkin matrix of (1) with natural boundary conditions, the reaction term mass-lumped (row sums), and $x_{M+1}, \dots, x_N$ the Dirichlet nodes. With the artificial diffusion matrix

$$d_{ij} = d_{ji} = -\max\{a_{ij}, 0, a_{ji}\} \quad (i \neq j), \qquad d_{ii} = -\textstyle\sum_{j\neq i} d_{ij}, \tag{2}$$

and the fluxes $f_{ij} = d_{ij}(u_j - u_i)$, the scheme reads [1, (11)]

$$\sum_{j=1}^{N} a_{ij}u_j + \sum_{j\neq i}(1-\alpha_{ij}(u))d_{ij}(u_j-u_i) = g_i, \quad i = 1,\dots,M, \qquad u_i = 0,\ i > M. \tag{3}$$

Limiter. Kuzmin's limiter as stated in [1, Sect. 4]: for $a_{ji} \le a_{ij}$ (node i "owns" the edge),

$$P_i^{\pm} = \sum_{j:\, a_{ji}\le a_{ij}} f_{ij}^{\pm}, \quad Q_i^{\pm} = -\sum_j f_{ij}^{\mp}, \quad R_i^{\pm} = \min\{1, Q_i^{\pm}/P_i^{\pm}\}, \quad \alpha_{ij} = \alpha_{ji} = \begin{cases} R_i^+, & f_{ij} > 0, \\ 1, & f_{ij} = 0, \\ R_i^-, & f_{ij} < 0, \end{cases}$$

with $R_i^{\pm} = 1$ when $P_i^{\pm} = 0$ and at Dirichlet nodes, so that $\alpha_{ij} = 1$ on edges owned by a Dirichlet node.

Variants. Table 1 uses the solution-independent weights $\alpha_{ij} = 0.5$ (with $\alpha_{ij} = 1$ on edges owned by a Dirichlet node, as in the paper); the problem is then linear. Table 11 replaces (2) by the convection matrix alone,

$$d_{ij} = -\max\{(\boldsymbol{b}\cdot\nabla\varphi_j, \varphi_i),\, 0,\, (\boldsymbol{b}\cdot\nabla\varphi_i, \varphi_j)\} \qquad (i \neq j), \tag{4}$$

which is [1, (54)].

Error measures. With $e_h = u - u_h$ and i_h the nodal interpolant, the tables report $\|e_h\|_0$, $|e_h|_1$,

$$d_h(u_h)^{1/2} := d_h(u_h; i_h u, i_h u)^{1/2}, \qquad d_h(w; v, v) = \sum_{i<j}(1-\alpha_{ij}(w))|d_{ij}|\,(v_i - v_j)^2,$$

and

$$\|e_h\|_h^2 = C_a(\varepsilon|e_h|_1^2 + \sigma_0\|e_h\|_0^2) + d_h(u_h; i_h u - u_h, i_h u - u_h), \qquad C_a = 1. \tag{5}$$

The paper defines $\|\cdot\|_h$ on the discrete space with an unspecified constant C_a; (5) is the reading that reproduces all its values (e.g. $\varepsilon = 10$, grid 1, level 3: $\sqrt{10}\,|e_h|_1 = 0.5556$ against $\|e_h\|_h = 0.5557$ printed).

3 Grids

Figure 1 shows the five grids on level 0.

- **Grid 1**: 4×4 squares, Friedrichs–Keller diagonals "/".
- **Grid 2**: 13 nodes and 16 triangles: the diamond through the edge midpoints and the square $[\frac{1}{4}, \frac{3}{4}]^2$ with both diagonals.

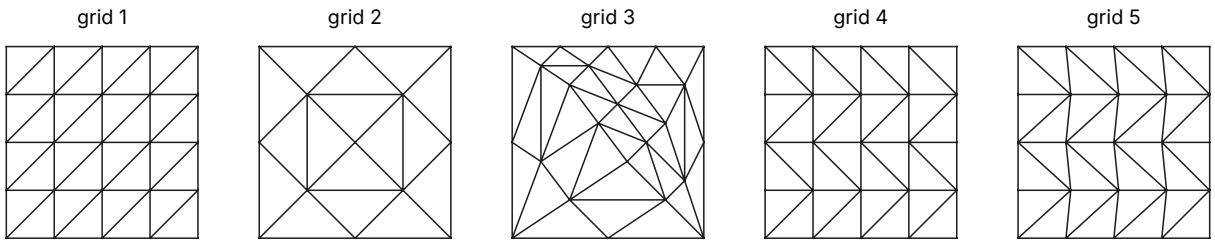


Figure 1: Grids 1–5 on level 0 as built by `run_bjk16.py`; compare [1, Fig. 1].

- **Grid 3**: unstructured, 24 nodes and 36 triangles. The paper gives no coordinates. Its Figure 1 is vector graphics (a `gnuplot` file `grid_irregular.eps`); reading the drawing commands from the PDF gives the triangle list exactly, and all nodes turn out to be multiples of 1/20. A random Delaunay grid of the same size, tried first, misses the paper's values by about 30 %; the extracted grid reproduces them.
- **Grid 4**: grid 1 with "\" diagonals in the even rows of squares (counted from below).
- **Grid 5**: grid 4 with the interior nodes of the even horizontal mesh lines shifted to the right by $h/10$, so that $\theta_{ij} > \pi/2$ for every diagonal edge (non-Delaunay).

Level l of grids 1, 4, 5 has $4 \cdot 2^l$ squares per side, $h = 2^{-l-2}$; grids 2 and 3 are refined uniformly (red refinement) l times. Level 8 has 1 050 625 nodes on grids 1, 4, 5.

4 Nonlinear solvers

The paper solves (3) by a fixed-point iteration with Anderson acceleration [5] and stops when the Euclidean norm of the residual is below 10^{-9}. We use two solvers, both stopped at 10^{-10}.

- **Semismooth Newton** with the exact generalised Jacobian of the limiter on its current smooth branch and an Armijo line search (default).
- **Anderson-accelerated fixed point** (`fp_anderson`): the map $u \mapsto \tilde{u}$, $(A + D)\tilde{u} = g - L((1 - \alpha(u))d - d)u$, where $L(w)$ is the graph Laplacian with edge weights w, so that one LU factorisation of $A + D$ serves all iterations; Anderson mixing of depth 10 in the form of [5, Sect. 3]; if the best residual has not halved within 200 iterations, the history is cleared and the iteration restarts from the best iterate.

Both give the same values to all printed digits wherever both converge. Their costs differ sharply:

- For $\varepsilon = 10$ and for $\varepsilon = 10^{-8}$ on grids 1 and 2, Newton needs 1–42 iterations on every level.
- For $\varepsilon = 10^{-8}$ on grids 3–5 the number of Newton steps grows quickly (grid 4: 22, 34, 233 on levels 4, 5, 6), and on grid 4 level 7 Newton had not converged after four hours. Anderson solves grids 4 and 5 on level 7 in 5122 and 4984 iterations (21 and 20 minutes), and on level 8 in about 7000 iterations (under three hours).
- On **grid 3, level 7**, Anderson converges linearly to a residual of $1.4 \cdot 10^{-8}$ and then stagnates; clearing the history does not help, and Newton started from that iterate only takes steps of length 10^{-3}–10^{-4}. The stagnation is therefore a property of the nonlinear problem (switching

of the limiter at near-ties of nodal values) and not of the acceleration. The row of Table 4 for level 7 is the Anderson iterate after 3400 iterations, residual $1.35 \cdot 10^{-8}$. Its errors agree with the paper in all printed digits except $\|e_h\|_h = 9.6499 \cdot 10^{-4}$ against $9.649 \cdot 10^{-4}$.

5 Results

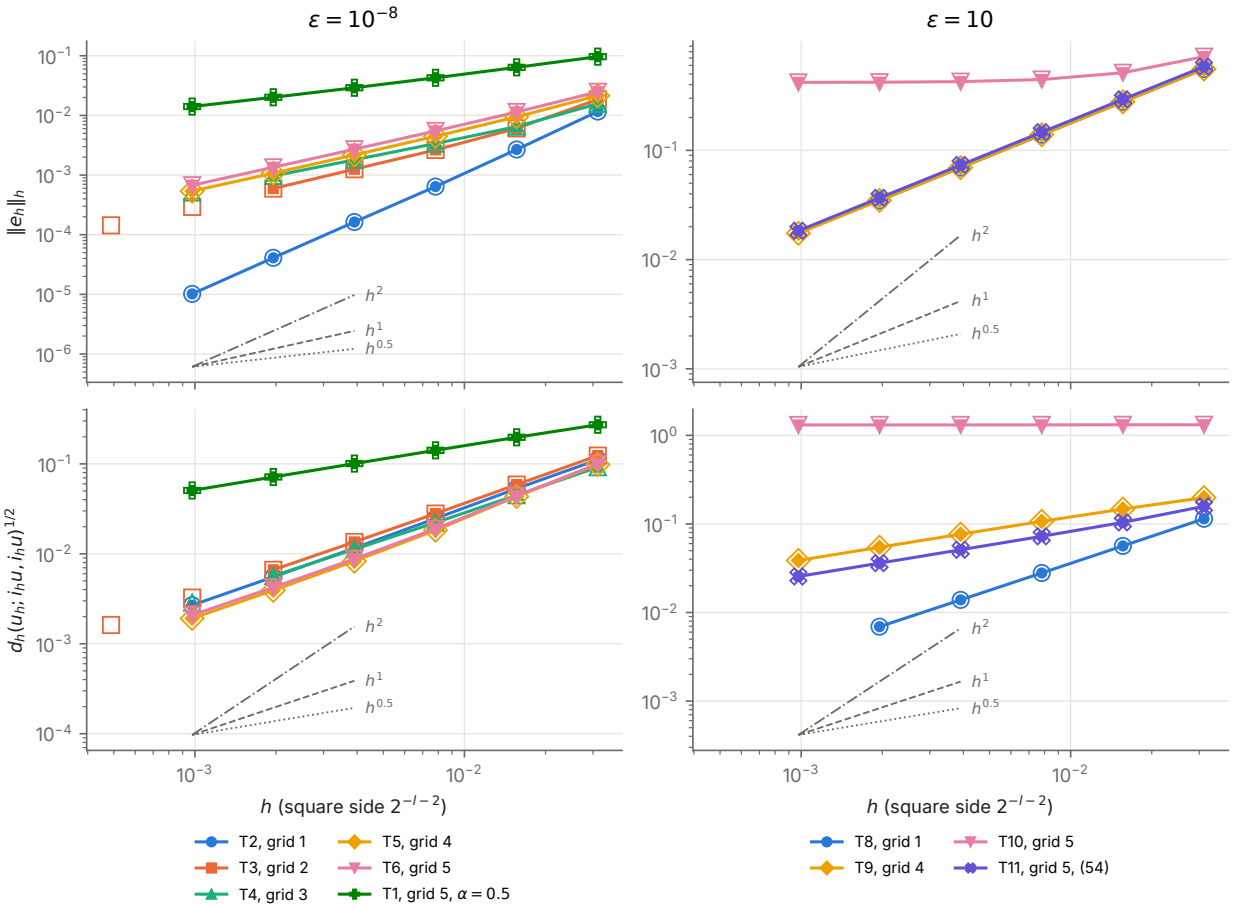


Figure 2: $\|e_h\|_h$ (top) and $d_h(u_h; i_h u, i_h u)^{1/2}$ (bottom) against h, for $\varepsilon = 10^{-8}$ (left) and $\varepsilon = 10$ (right). Lines with filled markers: galerkit; open markers: the values printed in the paper (Table 3 also has levels 8 and 9, not recomputed); grey: reference slopes. The colour follows the grid.

Tables 1–11 carry the numbers of the paper's tables and repeat their layout, with the galerkit values (Table S is the summary above). Entries that differ from the paper in a printed digit are in colour, and the caption gives the paper's value.

Table 1: $\varepsilon = 10^{-8}$, grid 5, constant weights $\alpha_{ij} = 0.5$. galerkit values; "ord." is $\log_2$ of the ratio to the previous level. Every entry agrees with the paper in all printed digits.

l	$\|e_h\|_0$	ord.	$\|e_h\|_1$	ord.	$d_h(u_h)^{1/2}$	ord.	$\|e_h\|_h$	ord.
3	2.622e−2		7.668e−1		2.722e−1		9.666e−2	
4	1.527e−2	0.78	7.021e−1	0.13	1.975e−1	0.46	6.397e−2	0.60
5	8.260e−3	0.89	6.489e−1	0.11	1.415e−1	0.48	4.274e−2	0.58
6	4.295e−3	0.94	6.149e−1	0.08	1.008e−1	0.49	2.912e−2	0.55
7	2.189e−3	0.97	5.956e−1	0.05	7.150e−2	0.49	2.015e−2	0.53
8	1.105e−3	0.99	5.854e−1	0.03	5.065e−2	0.50	1.408e−2	0.52

Table 2: $\varepsilon = 10^{-8}$, grid 1. galerkit values; "ord." is $\log_2$ of the ratio to the previous level. Entries in colour differ from the paper in the printed digits; the paper has $\|e_h\|_h$ at $l = 7$: 4.099e−5, $\|e_h\|_0$ at $l = 8$: 5.357e−6, $|e_h|_1$ at $l = 8$: 6.822e−3, $\|e_h\|_h$ at $l = 8$: 1.018e−5.

l	$\|e_h\|_0$	ord.	$\|e_h\|_1$	ord.	$d_h(u_h)^{1/2}$	ord.	$\|e_h\|_h$	ord.
3	5.457e−3		2.287e−1		1.112e−1		1.163e−2	
4	1.408e−3	1.95	1.074e−1	1.09	5.317e−2	1.06	2.683e−3	2.12
5	3.493e−4	2.01	5.113e−2	1.07	2.472e−2	1.11	6.410e−4	2.07
6	8.652e−5	2.01	2.546e−2	1.01	1.158e−2	1.09	1.633e−4	1.97
7	2.152e−5	2.01	1.321e−2	0.95	5.533e−3	1.07	4.100e−5	1.99
8	5.358e−6	2.01	6.827e−3	0.95	2.685e−3	1.04	1.017e−5	2.01

Table 3: $\varepsilon = 10^{-8}$, grid 2. galerkit values; "ord." is $\log_2$ of the ratio to the previous level. Every entry agrees with the paper in all printed digits.

l	$\|e_h\|_0$	ord.	$\|e_h\|_1$	ord.	$d_h(u_h)^{1/2}$	ord.	$\|e_h\|_h$	ord.
3	8.533e−3		2.901e−1		1.236e−1		1.855e−2	
4	2.516e−3	1.76	1.954e−1	0.57	5.884e−2	1.07	6.065e−3	1.61
5	8.369e−4	1.59	1.380e−1	0.50	2.801e−2	1.07	2.640e−3	1.20
6	2.891e−4	1.53	1.031e−1	0.42	1.356e−2	1.05	1.254e−3	1.07
7	1.103e−4	1.39	7.865e−2	0.39	6.638e−3	1.03	5.938e−4	1.08

Table 4: $\varepsilon = 10^{-8}$, grid 3. galerkit values; "ord." is $\log_2$ of the ratio to the previous level. Entries in colour differ from the paper in the printed digits; the paper has $\|e_h\|_h$ at $l = 7$: 9.649e−4.

l	$\|e_h\|_0$	ord.	$\|e_h\|_1$	ord.	$d_h(u_h)^{1/2}$	ord.	$\|e_h\|_h$	ord.
3	6.125e−3		3.202e−1		9.189e−2		1.569e−2	
4	2.216e−3	1.47	2.244e−1	0.51	4.488e−2	1.03	6.502e−3	1.27
5	9.946e−4	1.16	1.821e−1	0.30	2.224e−2	1.01	3.376e−3	0.95
6	4.993e−4	0.99	1.559e−1	0.22	1.124e−2	0.98	1.802e−3	0.91
7	2.519e−4	0.99	1.375e−1	0.18	5.676e−3	0.99	9.650e−4	0.90

Table 5: $\varepsilon = 10^{-8}$, grid 4. galerkit values; "ord." is $\log_2$ of the ratio to the previous level. Entries in colour differ from the paper in the printed digits; the paper has $\|e_h\|_h$ at $l = 8$: 5.381e−4.

l	$\|e_h\|_0$	ord.	$\|e_h\|_1$	ord.	$d_h(u_h)^{1/2}$	ord.	$\|e_h\|_h$	ord.
3	6.383e−3		4.826e−1		9.814e−2		2.143e−2	
4	2.313e−3	1.46	4.543e−1	0.09	4.341e−2	1.18	9.455e−3	1.18
5	1.089e−3	1.09	4.434e−1	0.03	1.830e−2	1.25	4.469e−3	1.08
6	5.527e−4	0.98	4.361e−1	0.02	8.276e−3	1.14	2.176e−3	1.04
7	2.817e−4	0.97	4.320e−1	0.01	3.926e−3	1.08	1.077e−3	1.01
8	1.425e−4	0.98	4.297e−1	0.01	1.915e−3	1.04	5.380e−4	1.00

Table 6: $\varepsilon = 10^{-8}$, grid 5. galerkit values; "ord." is $\log_2$ of the ratio to the previous level. Entries in colour differ from the paper in the printed digits; the paper has $\|e_h\|_h$ at $l = 8$: 6.755e−4.

l	$\|e_h\|_0$	ord.	$\|e_h\|_1$	ord.	$d_h(u_h)^{1/2}$	ord.	$\|e_h\|_h$	ord.
3	6.925e−3		5.638e−1		9.992e−2		2.486e−2	
4	2.687e−3	1.37	5.395e−1	0.06	4.405e−2	1.18	1.140e−2	1.12
5	1.304e−3	1.04	5.294e−1	0.03	1.896e−2	1.22	5.491e−3	1.05
6	6.645e−4	0.97	5.225e−1	0.02	8.792e−3	1.11	2.711e−3	1.02
7	3.382e−4	0.97	5.186e−1	0.01	4.235e−3	1.05	1.349e−3	1.01
8	1.708e−4	0.99	5.164e−1	0.01	2.083e−3	1.02	6.754e−4	1.00

Table 7: $\varepsilon = 10^{-8}$, grid 1, arithmetic mean of $1 - \alpha_{ij}(u_h)$ over all edges.

l	galerkit	paper
3	1.09e-01	1.09e-01
4	5.94e-02	5.94e-02
5	3.16e-02	3.16e-02
6	1.73e-02	1.73e-02
7	9.60e-03	9.60e-03
8	5.27e-03	5.27e-03

Table 8: $\varepsilon = 10$, grid 1. galerkit values; "ord." is $\log_2$ of the ratio to the previous level. Every entry agrees with the paper in all printed digits.

l	$\|e_h\|_0$	ord.	$\|e_h\|_1$	ord.	$d_h(u_h)^{1/2}$	ord.	$\|e_h\|_h$	ord.
3	2.148e−3		1.757e−1		1.144e−1		5.557e−1	
4	5.379e−4	2.00	8.799e−2	1.00	5.643e−2	1.02	2.783e−1	1.00
5	1.345e−4	2.00	4.401e−2	1.00	2.792e−2	1.02	1.392e−1	1.00
6	3.360e−5	2.00	2.201e−2	1.00	1.387e−2	1.01	6.960e−2	1.00
7	8.398e−6	2.00	1.100e−2	1.00	6.912e−3	1.00	3.480e−2	1.00

Table 9: $\varepsilon = 10$, grid 4. galerkit values; "ord." is $\log_2$ of the ratio to the previous level. Every entry agrees with the paper in all printed digits.

l	$\|e_h\|_0$	ord.	$\|e_h\|_1$	ord.	$d_h(u_h)^{1/2}$	ord.	$\|e_h\|_h$	ord.
3	2.187e−3		1.756e−1		1.983e−1		5.554e−1	
4	6.209e−4	1.82	8.800e−2	1.00	1.473e−1	0.43	2.783e−1	1.00
5	1.940e−4	1.68	4.402e−2	1.00	1.069e−1	0.46	1.392e−1	1.00
6	6.899e−5	1.49	2.201e−2	1.00	7.657e−2	0.48	6.961e−2	1.00
7	2.789e−5	1.31	1.101e−2	1.00	5.450e−2	0.49	3.481e−2	1.00
8	1.239e−5	1.17	5.503e−3	1.00	3.867e−2	0.50	1.740e−2	1.00

Table 10: $\varepsilon = 10$, grid 5. galerkit values; "ord." is $\log_2$ of the ratio to the previous level. Every entry agrees with the paper in all printed digits.

l	$\|e_h\|_0$	ord.	$\|e_h\|_1$	ord.	$d_h(u_h)^{1/2}$	ord.	$\|e_h\|_h$	ord.
3	1.248e−2		2.229e−1		1.317e0		7.211e−1	
4	1.123e−2	0.15	1.558e−1	0.52	1.316e0	0.00	5.135e−1	0.49
5	1.090e−2	0.04	1.333e−1	0.22	1.313e0	0.00	4.452e−1	0.21
6	1.080e−2	0.01	1.269e−1	0.07	1.312e0	0.00	4.259e−1	0.06
7	1.077e−2	0.00	1.252e−1	0.02	1.311e0	0.00	4.207e−1	0.02
8	1.076e−2	0.00	1.248e−1	0.01	1.310e0	0.00	4.193e−1	0.00

Table 11: $\varepsilon = 10$, grid 5, d_{ij} from the convection matrix (4). galerkit values; "ord." is $\log_2$ of the ratio to the previous level. Every entry agrees with the paper in all printed digits.

l	$\|e_h\|_0$	ord.	$\|e_h\|_1$	ord.	$d_h(u_h)^{1/2}$	ord.	$\|e_h\|_h$	ord.
3	2.319e−3		1.849e−1		1.581e−1		5.846e−1	
4	6.098e−4	1.93	9.275e−2	1.00	1.040e−1	0.60	2.933e−1	1.00
5	1.676e−4	1.86	4.642e−2	1.00	7.244e−2	0.52	1.468e−1	1.00
6	4.979e−5	1.75	2.322e−2	1.00	5.105e−2	0.50	7.343e−2	1.00
7	1.659e−5	1.59	1.161e−2	1.00	3.607e−2	0.50	3.672e−2	1.00
8	6.302e−6	1.40	5.806e−3	1.00	2.550e−2	0.50	1.836e−2	1.00

6 Discussion

6.1 The seven deviations

All seven deviations are on levels 7 and 8, i.e. on $2.6 \cdot 10^5$–$1.05 \cdot 10^6$ unknowns, in the convection-dominated regime, where the limiter is active on a large set of edges. Six are one unit in the fourth digit; the largest is $|e_h|_1$ on grid 1, level 8 ($6.827 \cdot 10^{-3}$ against $6.822 \cdot 10^{-3}$, 0.07 %). None appears for the linear problem of Table 1 or for $\varepsilon = 10$, where the nonlinear residual is driven to 10^{-11}–10^{-16}. Two differences in the setup can account for them: the paper stopped at a Euclidean residual of 10^{-9} (ours: 10^{-10}), and [1] proves existence but not uniqueness of solutions of the nonlinear problem, so the digits beyond the stopping tolerance need not be determined by the scheme alone. We did not rerun with the paper's tolerance, because the result of a fixed-point iteration stopped early depends on the whole iteration path, which the paper does not specify (start value, Anderson depth, damping).

6.2 Three choices made explicit

1. **Table 1 is computed on grid 5.** The caption of Table 1 says grid 5, the text of Section 8.1 says grid 1. On grid 5 every entry agrees; on grid 1 the level-3 values are $\|e_h\|_0 = 1.951 \cdot 10^{-2}$, $|e_h|_1 = 0.4408$, $d_h^{1/2} = 0.2528$, $\|e_h\|_h = 4.355 \cdot 10^{-2}$, against $2.622 \cdot 10^{-2}$, 0.7668, 0.2722, $9.666 \cdot 10^{-2}$ in the paper. The text is a typo.

2. **The mean in Table 7 is over all edges**, including the edges between Dirichlet nodes, on which $\alpha_{ij} = 1$. Over the edges with two free end points the level-3 mean is $1.04 \cdot 10^{-1}$ instead of the printed $1.09 \cdot 10^{-1}$.

3. **The error norm** is (5) with $C_a = 1$ and the stabilisation term at $i_h u - u_h$.

6.3 What the data say about the theory

Corollary 17 of [1] gives, for $\sigma_0 > 0$ and any symmetric limiter with values in $[0, 1]$,

$$\|u - u_h\|_h \le C\big(\varepsilon + \sigma_0^{-1}\{\|\boldsymbol{b}\|_{0,\infty}^2 + \|c\|_{0,\infty}^2\} + \sigma_0 h^2\big)^{1/2} h \|u\|_2 + C\big(\varepsilon + \|\boldsymbol{b}\|_{0,\infty} h\big)^{1/2} |i_h u|_1,$$

the second term coming from $d_h(u_h; i_h u, i_h u) \le C(\varepsilon + \|\boldsymbol{b}\|_{0,\infty} h)|i_h u|_1^2$ (Lemma 16). The tables test each ingredient.

- **Sharpness for general limiters.** With $\alpha_{ij} = 0.5$ (Table 1), $d_h^{1/2}$ and $\|e_h\|_h$ converge with orders 0.49–0.50 and 0.52–0.55 on the last levels: the $O(h^{1/2})$ of (52) and of Corollary 17 is attained, so the estimate cannot be improved under the hypotheses "$\alpha_{ij} \in [0, 1]$, $\alpha_{ij} = \alpha_{ji}$" alone. Note that $|e_h|_1$ does not converge in this case (0.59 on level 8), which the estimate allows, since $\varepsilon |e_h|_1^2$ is the only H^1 control in $\|\cdot\|_h$.

- **Kuzmin's limiter is better than the estimate.** On grid 1 (Table 2) $d_h^{1/2}$ converges with order ≈ 1 and $\|e_h\|_h$ with order 2; the mean of $1 - \alpha_{ij}$ (Table 7) decays like $h^{0.85}$–$h^{0.91}$, which quantifies how rough the bound $1 - \alpha_{ij} \le 1$ in Lemma 16 is. On grids 2–5, $d_h^{1/2}$ still converges with order ≈ 1, but $\|e_h\|_h$ only with order ≈ 1 and $|e_h|_1$ stagnates on grids 4 and 5 (order $\to 0$), so the gain over Corollary 17 is mesh dependent.

- **The diffusion-dominated regime follows Remark 18.** For $\varepsilon = 10$, $d_h^{1/2}$ converges with order 1 on grid 1 (Delaunay; Remark 18(a) predicts at least 1/2), with order 1/2 on grid 4

(right angles, $\theta_h = \pi/2$, Remark 18(c): $d_{ij} \neq 0$ persists), and not at all on grid 5 ($\theta_{ij} > \pi/2$, Remark 18(d)), where also $\|e_h\|_0$ and $|e_h|_1$ stagnate (Table 10). Building D from the convection matrix (4) restores order 1/2 for $d_h^{1/2}$ and order 1 for $\|e_h\|_h$ on grid 5 (Table 11), as the paper's estimate after (54) predicts; $\|e_h\|_0$ then converges with order decreasing from 1.9 to 1.4, i.e. the L^2 error is affected by the $O(h^{1/2})$ consistency error.

Minimality and the sharper track. The estimate of Corollary 17 uses only $\alpha_{ij} \in [0, 1]$ and symmetry; Table 1 shows that under these hypotheses the order 1/2 is optimal, so no hypothesis can be dropped without losing the estimate and none needs to be added to attain it. To gain more, a hypothesis on the limiter itself is required. The later literature follows exactly this track: the limiter of Barrenechea, John & Knobloch (M3AS 2017) [2] satisfies the discrete maximum principle and is linearity preserving on arbitrary simplicial meshes, and the monotone upwind-type schemes of John & Knobloch [4] are designed for the same purpose; the survey [3] collects the current state. The present data (Table 10 against Table 11) show the alternative cure proposed in [1] itself, a different matrix D, which repairs the non-Delaunay case at the price of order 1/2 for $d_h^{1/2}$ even in the diffusion-dominated regime. The galerkit reproduction of [2] (Tables 1–3, in `examples/convection_diffusion/2d/afc_linearity_preserving`) is the natural continuation.

7 Reproducibility

All files are in `examples/convection_diffusion/2d/afc_bjk16` of the galerkit repository (branch `feature/adaptive-afc-3d`).

- `run_bjk16.py`: grids, cases and the runs; results to `output/t*.json`.

```
python run_bjk16.py                                    # Tables 1-11, levels 3-7
python run_bjk16.py t1 t9 t10 t11 --levels 8           # Newton, 5-10 min each
python run_bjk16.py t2 t5 t6 --levels 8 --solver fp_anderson --maxiter 200000
```

- `paper_values.py`: the values printed in the paper; `compare.py`: the comparison (README section); `make_report.py`: the tables and figures of this report; `results.json`: the computed values used here.

- Library additions (`src/galerkit/convection_diffusion/afc.py`): solution-independent weights (`alpha_fixed`), d_{ij} from another matrix (`d_matrix`), and the Anderson-accelerated fixed point (`afc_solver="fp_anderson"`). The test `tests/test_cdr_afc.py` checks Tables 1, 2 and 11 on level 3 and the agreement of the Anderson and Newton solutions.

References

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[2] G. R. Barrenechea, V. John, P. Knobloch, An algebraic flux correction scheme satisfying the discrete maximum principle and linearity preservation on general meshes, *Math. Models Methods Appl. Sci.* 27 (2017) 525–548. https://doi.org/10.1142/S0218202517500087

[3] G. R. Barrenechea, V. John, P. Knobloch, Finite element methods respecting the discrete maximum principle for convection-diffusion equations, *SIAM Rev.* 66 (2024) 3–88.

[4] V. John, P. Knobloch, On algebraically stabilized schemes for convection–diffusion–reaction problems, *Numer. Math.* 152 (2022) 553–585.

[5] H. F. Walker, P. Ni, Anderson acceleration for fixed-point iterations, *SIAM J. Numer. Anal.* 49 (2011) 1715–1735.